



SKG14 IC8 Pacing & Emerging Technologies for ICM Engineers L2

14.0 Remote Operation & Maintenance

July 2025

The PETRONAS Group adopts zero tolerance against all forms of bribery and corruption. We abide by the PETRONAS Code of Conduct and Business Ethics (CoBE) & Anti-Bribery and Corruption (ABC) Manual, guided by our Shared Values and Statement of Purpose.

© 2025 Petroliaam Nasional Berhad (PETRONAS)

All rights reserved. No part of this document may be reproduced in any form possible, stored in a retrieval system, transmitted and/or disseminated in any form or by any means (digital, mechanical, hard copy, recording or otherwise) without the permission of the copyright owner.

Course Content

14.1	Case for Change – Why Remote Autonomous Operation (RAO) ? New way of working through RAO	07
14.2	RAO definition and Level of Autonomy Scope and human's role in Operation	13
14.3	RAO Technical Requirements – Instrument, Control System and Telecommunications Technical and standards requirements	20
14.4	Control System Design Extension Options How do we extend the local site to remote site ?	25
14.5	Remote Operation Centre (ROC) Design Considerations Technical requirements for ROC	34
14.6	Remote Access, Remote Maintenance and Analytics New way of working in maintenance	37
14.7	Advantages and Challenges of Remote Autonomous Operation The triumph and trials!	43
14.8	RAO in supporting Sustainability How do we support MFT 0 ?	46

Learning Outcome

At the end of the session, you will be able to:

- Explain the different levels of autonomy
- Explain the definition and terminologies used in remote operation
- Describe the remote operation design and remote operation centre (ROC) technical requirements
- Describe the remote operation control system extension options
- Describe use cases & application of remote operation for Operation & Maintenance

SKG-14 Technology Inventory & Ruler

Pacing 4: Remote Operations & Maintenance

SKG 14 TI&R for Pacing 4 - Remote Operations & Maintenance provides the minimum competencies requirement for Instrument & Control fraternity for managing remote operations & maintenance from Instrument perspective.

7-3-13-0003-QR2 v3.0 SKG 14 – BASE Technology Competency Matrix / Descriptor Discipline – Instrument & Control Sub-Discipline – Instrument & Control Effective Date – 01 July 2025					
TECHNOLOGY INVENTORY		COMPETENCE LEVELS			
		Awareness	Knowledge	Skill	Expert
P4 Remote Operations & Maintenance The technology to safely and reliably operate a process facility and perform diagnostics, troubleshooting and maintenance from an external remote location with a high level of autonomy. Autonomy is the ability of a system to achieve operational goals in complex domains by making decisions and executing actions on behalf of or in cooperation with humans. The key enabling technologies are: a. Low latency, high reliability, high bandwidth communications b. Smart sensors/equipment c. Remote sensing d. Artificial Intelligence (AI) and Machine Learning (ML) e. Digital reality f. Robotics and drones Specify the requirements of the above technologies in specific applications to achieve high autonomy systems. Specify system (BPCS, SIS, packaged systems, communication etc) architecture and security requirements for remote operation in compliance to PETRONAS Cybersecurity governance.  Health, Safety & Environment (HSE)  Sustainability  Digitalization  Commercialization	Explain the different levels of autonomy. Aware of the advantages and challenges of a highly autonomous remote operations and maintenance. Basic understanding of common terminologies (e.g. latency, bandwidth) and technologies for communication (e.g. fiber optics, Digital Microwave Radio (DMR), VSAT, Low-Earth Orbit (LEO) satellites, pervasive WiFi, 4G/5G, backbone, etc). Aware of use cases of the listed technologies in remote operations and maintenance functions.	Describe specific remote operation centre design considerations (e.g. remote console operation, remote ESD function, site surveillance, emergency response).  Describe the advantages and disadvantages between control system extension options and its architecture.  Explain the common terminologies and technologies for communication to support remote operations. Describe a use case for remote operations or maintenance, including typical requirements of the technology involved. 	Specify the implementation of remote operation & maintenance.  Consequently, able to demonstrate: - Architecture of implementation - Design and security requirements - Technology used to achieve high autonomy level - Multidiscipline work process requirements Specify the communication requirements to support remote operations. Specify the analytics tools required to support remote maintenance.	Challenge / evaluate the specifications and lead the implementation of a remote operation facility with high autonomy level.	

14.1 Case for Change - Why Remote Autonomous Operation (RAO) ?

New Way of Working through Remote Autonomous Operation (RAO)

Case for Change: Oil & Gas industry experience significant challenges

Location

Hydrocarbon reserves located in harsh or remote location, therefore challenging and expensive to develop.

Geopolitical challenges.



Complex Plant Feed

Safety Risk,
Hazardous
Operations Aging
Asset

e.g.. High CO2 and
H2S, Emulsion, Sand.
Require development
of complex
processing
module/train.



Resource Optimization

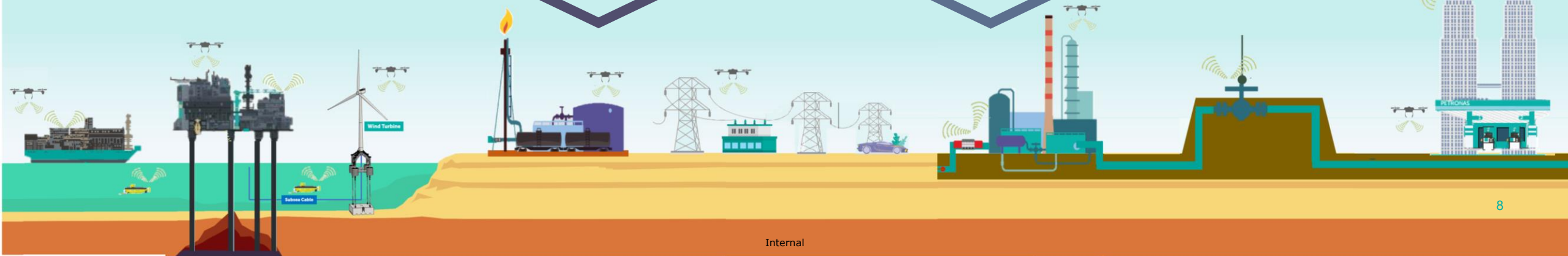
OPEX cost remain
unchanged at Business
OPU.



Pandemic & Market Volatility

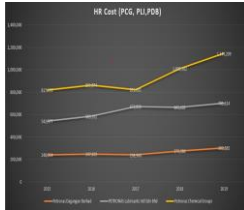
Covid-19 create
operational
disruption and
supply chain risk.

Rapid fluctuation in
global chemical
industry. Margins &
market becoming
more competitive.

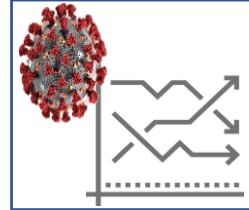


Oil and Gas companies to undertake Cost Optimization efforts to remain viable

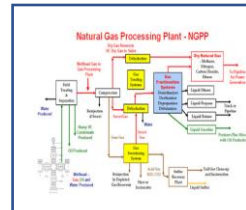
Current Scenarios



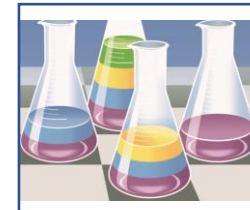
**OPEX Cost
Remain Unchanged**



**Black Swan Events
and Volatile Markets**



**Safety Risk,
Hazardous
Operations,
Aging Asset**



**More complex feed,
Product properties**



**Geopolitical
Challenges**

How does Industry respond?



ADNOC Panorama Command Center:

1. Business Value Chain link to Plant Operation workflow
2. Faster Decision Making
3. Centralized resources & support for local operation
4. Year 2017



Saudi Aramco:

1. Command Center
2. Year 2010
3. Riyadh, Saudi Arabia

Air Liquide:

1. Smart Innovative Operation Center, Kuala Lumpur
2. Year 2018
3. South East Asia Control



Shell Bridge

1. Proactive Technical Monitoring System
2. Year 2008
3. Gulf of Mexico

SHELL BRIDGE: EQUIPMENT SURVEILLANCE



Case for Change: Oil & Gas industry experience significant challenges

Now...what should be OUR response?

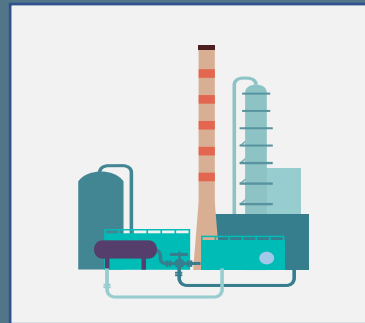


PETRONAS embarking change, finding solutions through optimization initiatives across its business.



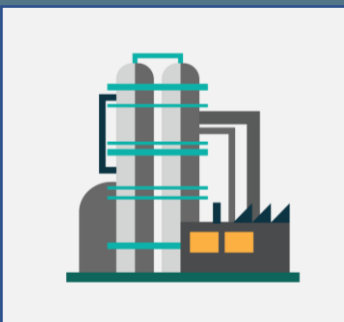
Upstream

1. EAGLE (Efficient & Agile) to Remote Ops offshore & offshore facilities, i.e. Resak, Dulang, Angsi.
2. Value Creation by reduction of OPEX at logistics, safety & maintenance (interval extension and analytic)



Gas New Energy (GNE)

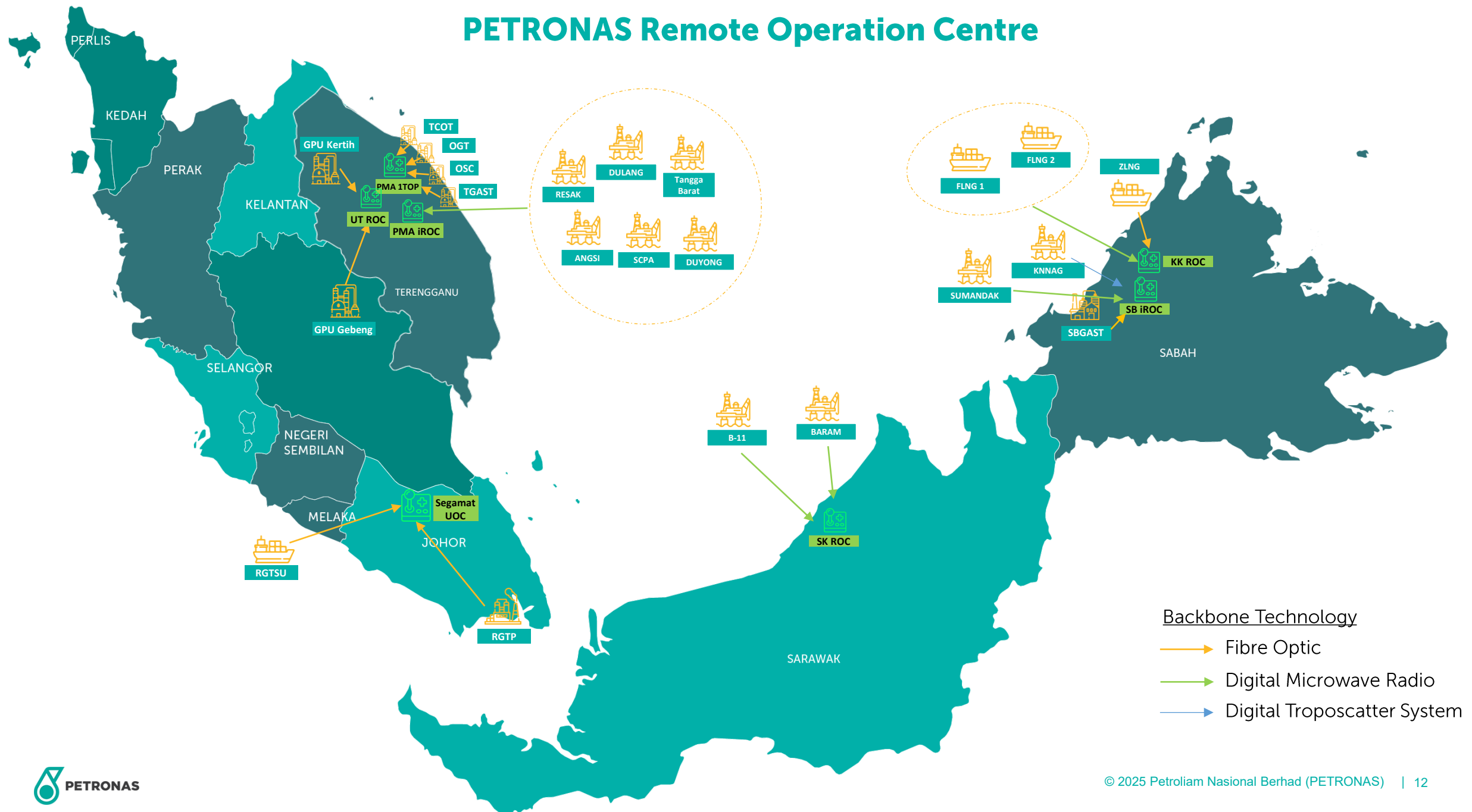
1. CORE (Centralize Operation, Remote Engineering), RGTSU, FLNG, GPU, PGB ROC.
2. Value Creation by reduction of OPEX at maintenance (interval extension and analytic)



Downstream (PCG, Refinery)

1. ROI via e.g. Optimization of Operation and Maintenance
2. EVP aspiration Remote Operations for new asset

PETRONAS Remote Operation Centre



14.2 RAO Definition and Level of Autonomy

Scope and human's role in Operation

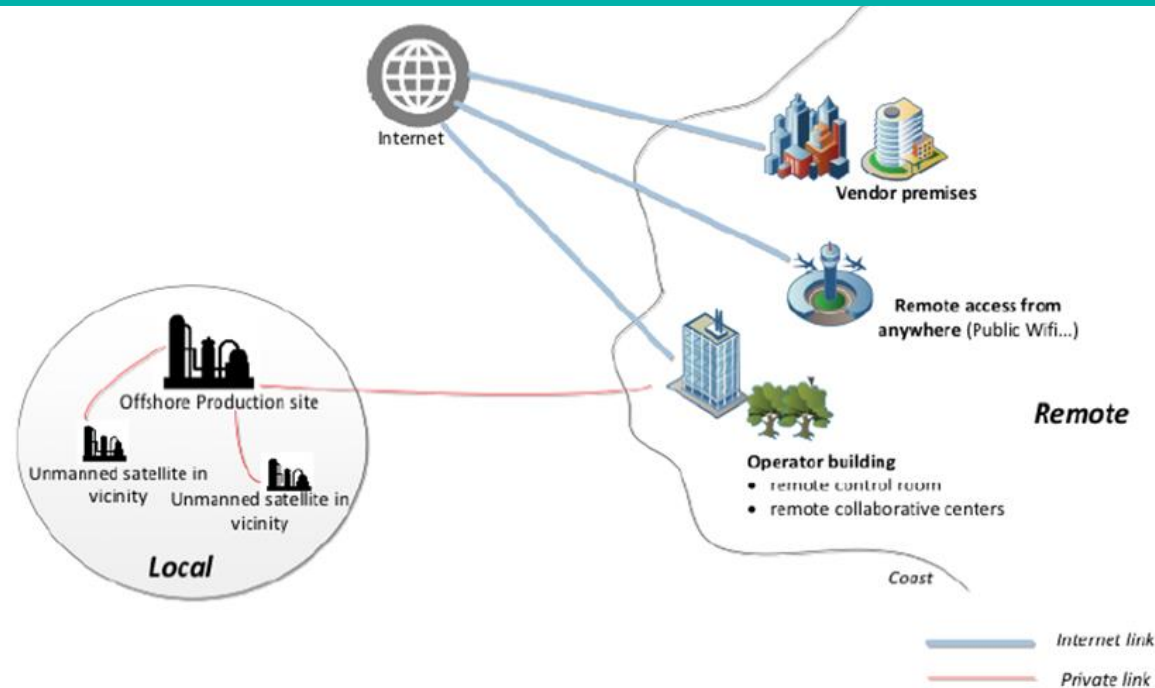
Definition of local and remote used in remote operation



Local – the process or production facility i.e., offshore platform, onshore terminals, onshore Plants



Remote - distant location away from production site where the new remote operation center will be located



These are the definitions of terms used in remote operation initiative



Remote Operation

The ability to perform monitoring, control, operate and safeguarding of a production facility, from a Remote location that is far away from the process or production facility.



Remote Monitoring

The ability of ROC to remotely monitor, with view-only access, process value of the local process/production facility.



Remote Maintenance

The ability to perform engineering and maintenance activity via access to the equipment at local production site remotely. Additionally, able to perform analytics for predictive and prescriptive maintenance.



Remote Shutdown

The ability to trigger shutdown on the local production facility in event of emergency or abnormal situation.



Remote Operation Centre

A facility where Remote Operation activity is performed by Panel Operator. It can house the Operation team and other relevant stakeholders.

These are the technical terminologies that are used in remote operation technical design



Latency

The delay between sending a request and receiving a response, affecting how quickly data moves.

High latency leads to noticeable delays, while low latency ensures smoother and faster performance.



Reliability

Reliability in networking refers to the ability of a network to perform its required functions under given conditions for a designated period of time without failure. It ensures that network services and resources are available to users when needed, without interruptions



Bandwidth

The amount of information that can be sent or received over a network in a certain time. It can also mean the capacity for data transfer of an electronic communications system, such as the maximum data transfer rate of a network connection.



Autonomous / Autonomy

In the context of process industries, refers to systems and machinery that can perform tasks and make decisions independently, without requiring direct human intervention for every action.

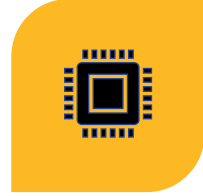
Involves use of advanced technologies, e.g., Artificial Intelligence (AI) and Machine Learning (ML) to enable self-learning and adaptation to new data or environments

These are the telecommunication backbone technologies that are used in remote operation to enable the communication from local site to ROC



FIBER OPTICS (FO)

- An assembly similar to an electrical cable but containing one or more optical fibers, used to carry light
- Optical fiber elements are individually coated with plastic layers and contained in a protective tube suitable for their environment
- Used for transmitting data, voice, and images by the passage of light.



DIGITAL MICROWAVE RADIO (DMR)

- Used for transmitting digital information via microwave or radio frequency (RF) signals.
- It involves use of microwave frequencies, ranging from 3-30 GHz, to carry digital data over long distances.
- Often used for point-to-point links, i.e., terminal at one end of the link communicates with a complementary terminal at the other end.



VERY SMALL APERTURE TERMINAL (VSAT)

- A two-way satellite ground station with a dish antenna <3.8 meters. These antennas range between 75cm-1.2m.
- These terminals are used to transmit and receive data from satellites, providing communication services such as satellite internet access, VOIP, and video.
- Commonly used in remote locations where traditional communication infrastructure is not available.



LOW EARTH ORBIT SATELLITE (LEO)

- Satellite that orbits the earth at an altitude between 160-2,000 km above the earth's surface, which allows them to provide high-speed, low-latency communication services.
- Ideal for applications such as satellite internet, remote sensing, and earth observation.
- Due to lower altitudes, they experience less signal delay compared to satellites in higher orbits, which is beneficial for real-time communication and data transfer



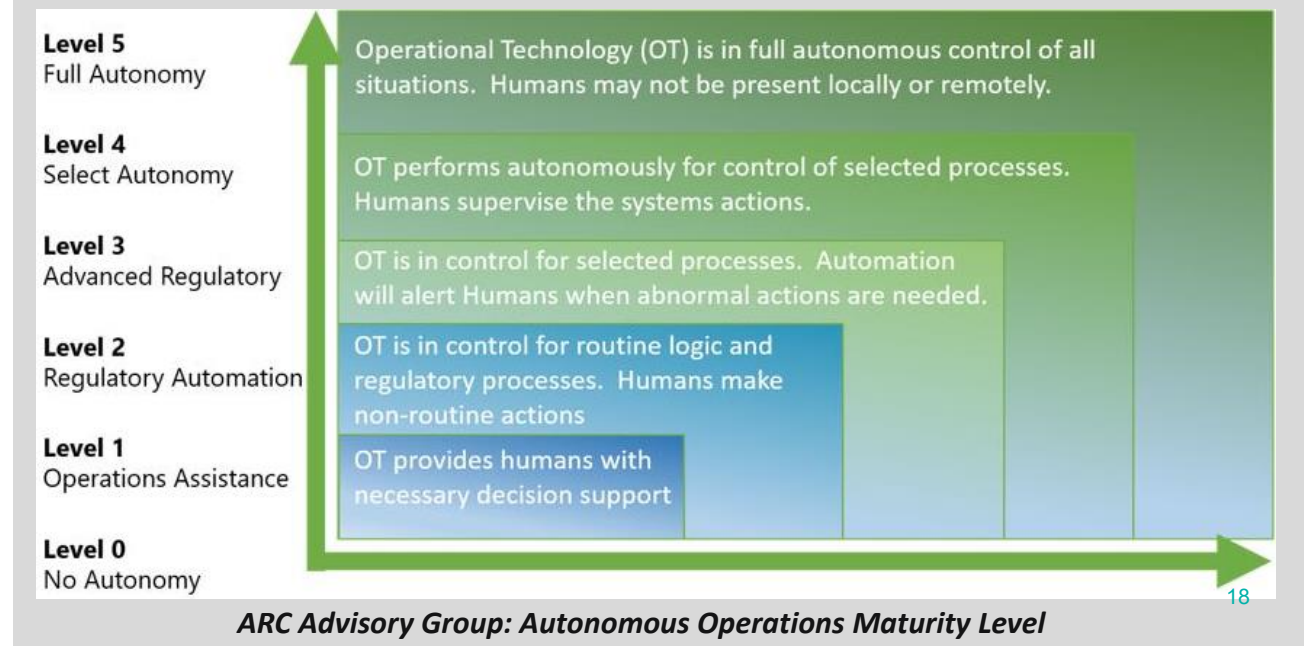
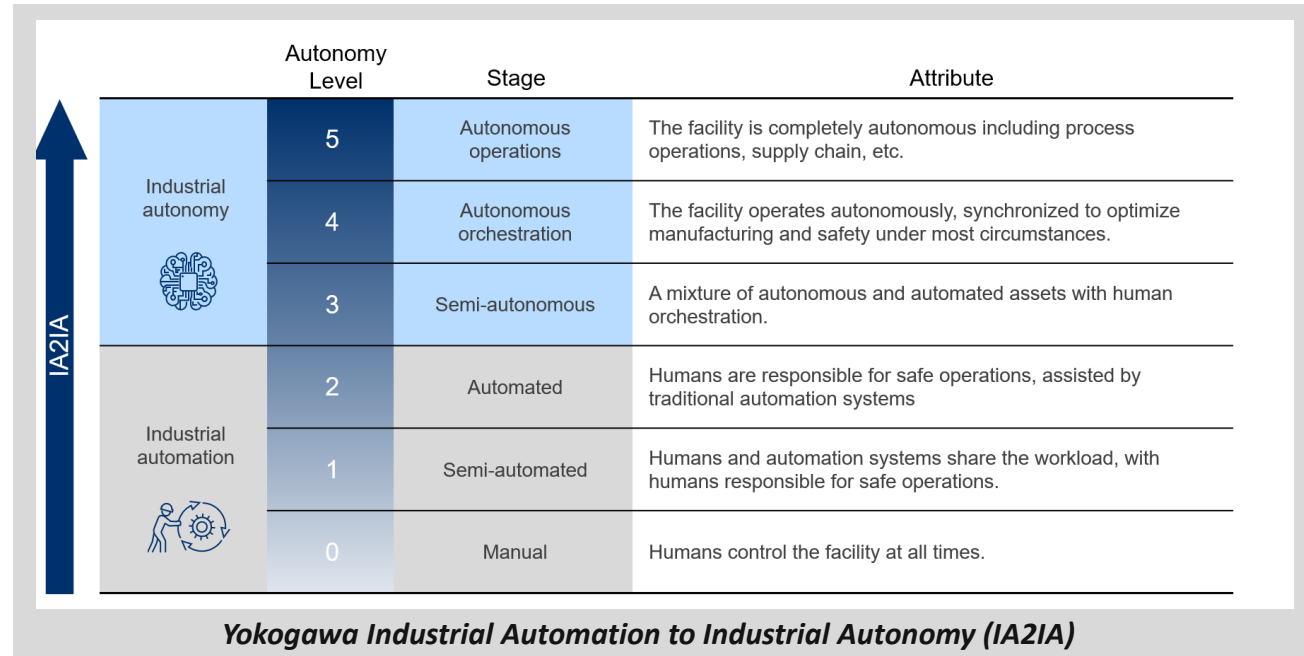
INDUSTRIAL WIRELESS

- Wireless communication technologies in industrial settings to connect devices, machines, and systems.
- Enables secure and reliable wireless connections for various industrial applications, such as manufacturing, process automation, and remote monitoring.
- Common types of industrial wireless technologies: pervasive wireless (Wireless Ethernet IEEE802.11), LoRaWAN, Bluetooth Low Energy, 5G

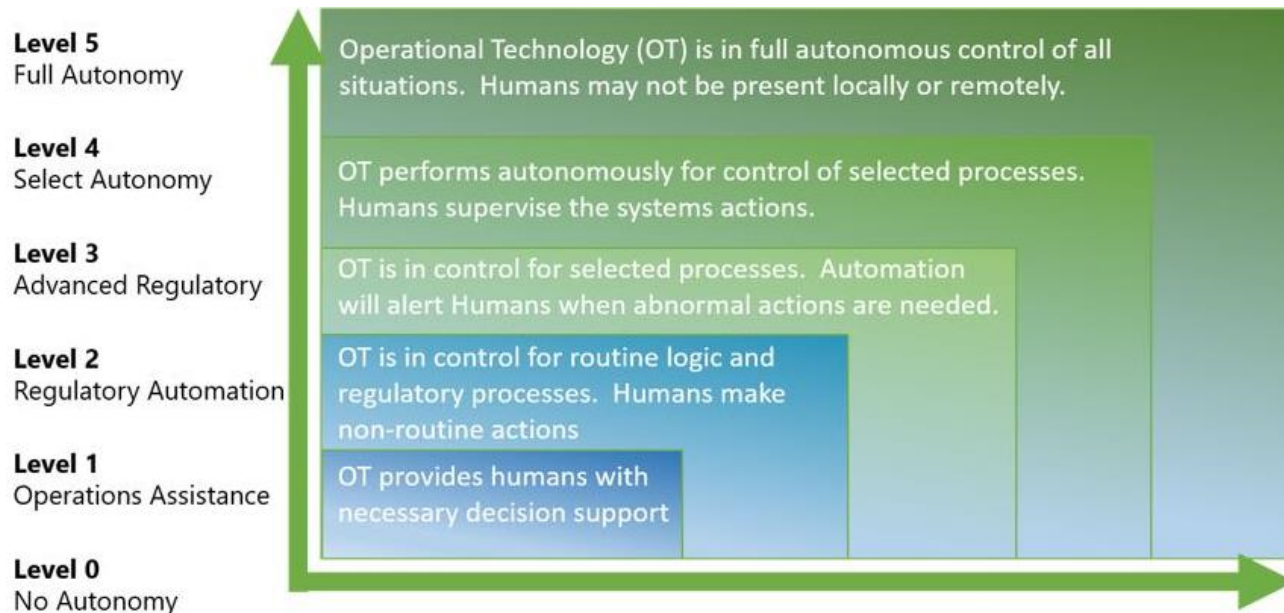
These are the autonomy levels defined in the industry

	Level 0	Level 1	Level 2	Level 3	Level 4	Level 5
Autonomous Plant	No Autonomy. Humans carry out all necessary operations without assistance.	Operations Assistance: Automation system provides decision support for necessary operations by remote / digital assistance. Human always responsible.	Automation system is in control in certain situations on request (humans pull support e.g., for plant start-up). Human always responsible.	Automation system is in control in certain situations. Plant actively alerts to issues and proposes solutions. Humans confirm.	Autonomous operations in certain situations: automation system has full control in these situations, humans supervise actions.	Fully autonomous operation in all situations. Humans may be completely absent.
Control Room Operation	Humans carry out control of all assets without assistance. Low level automation might be available e.g. safety systems	Automation of control loops during steady state. Manual start-up and shutdown of the plant. Manual execution of transitions. Alarm based notification.	Automation system assisted plant start-up, transition, steady state and shutdown. Manual fault correction supported by decision support system	Automated plant shutdown start-up and transition. On human request. Automatic correction of known deviations. Decision support for unexpected / unknown faults	Autonomous control in certain situations with automatic fault and deviation correction and avoidance	Full autonomous control, fault correction and avoidance in a situation. No human supervision required
Field Operation / Maintenance	All field operator tasks are executed by humans.	Automation system notifies humans about field activities e.g. operating valves	System guided field operation tasks. Humans get instructions what to do and when by decision support system	Most tasks required for standard operations are automated, like shutdown, start-up and transition phases. Number of humans in the field heavily reduced.	Almost human free field operation. Only human field operation in exceptional situations	Full autonomous field operations, no manual actions in the field necessary. No humans remain in the plant
Planning & Scheduling	Manual development of plans and the corresponding schedule	ERP plan creation on human request. Human decides when and how to execute the plans and adapt plans	Adaptations of plans to current situations by operator request	Continuous feedback and re-planning in case of production deviations. Manual schedule release.	Continuous autonomous planning and scheduling without user interaction. Detection of production deviations and re-planning. Manual schedule release.	Autonomous development and execution of plans and schedules. Autonomous re-planning in case of production deviations. No human interaction

International Federation of Automatic Control (IFAC): The Autonomous Industrial Plant



With increased autonomy, lower human intervention is required, increasing efficiency



Source: <https://www.arcweb.com/industry-best-practices/what-autonomous-operations>

- There are 6 levels of autonomy, starting from totally controlled by human to human completely absent
- Progression of the levels is dependent on the utilization of regulatory, advanced and AI/ML applied control systems with the intent to increase operation autonomy
- The degree of automation represents the degree of decision making (authority) deferred from the human to the system.

14.3 RAO Technical Requirements

Instrument, Control System and
Telecommunications

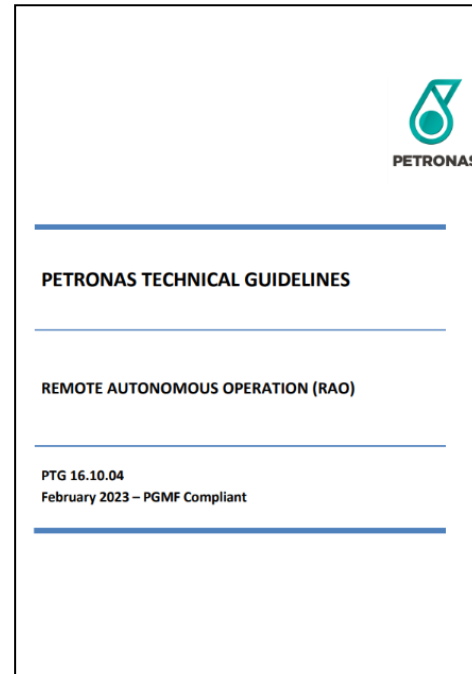
Technical and standards requirements

Remote Autonomous Operation relevant guidelines



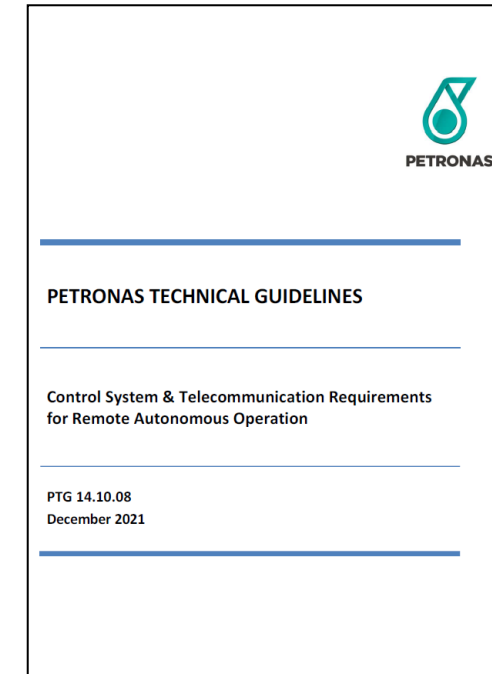
IOGP Report 627

Selection of system and security architectures for remote control, engineering, maintenance and monitoring.



PTG 16.10.04

Remote Autonomous Operation (RAO)

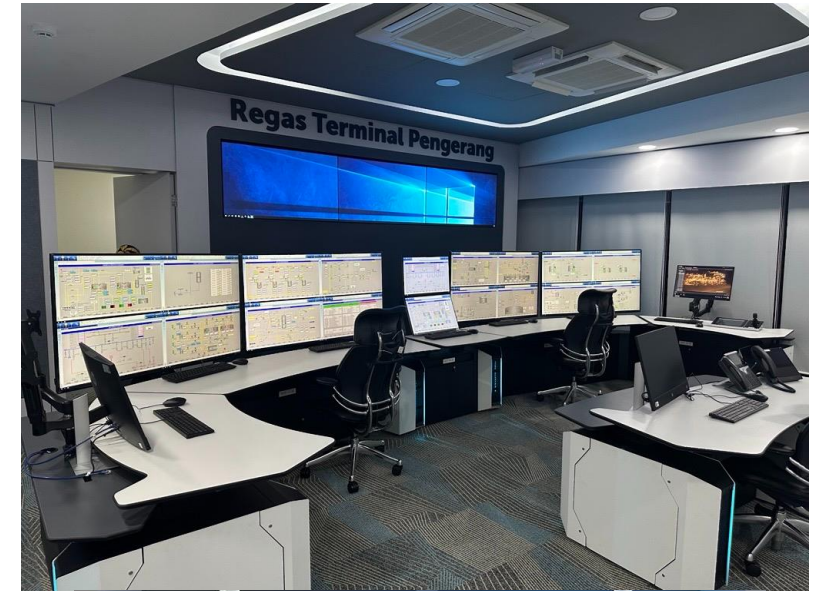


PTG 14.10.08

Control System & Telecommunication Requirements for Remote Autonomous Operation

Instrument & Control Technical Requirements

- Control system from local site to be extended to remote location
 - ✓ All subsystems, i.e., package equipment shall be connected to the DCS to enable single window of operation
 - ✓ The control system shall have read/write capability for all subsystems to enable control of the subsystem through DCS
- Remote Operation Center shall be provisioned to house the extended control system and the necessary equipment, e.g., Operator Workstation
- ROC shall have remote shutdown pushbutton(s) that is easily accessible to Panel Operator.
 - ✓ Loss of communication between ROC and local facility shall not result in tripping
 - ✓ Communication link shall be non-fail-safe design to avoid spurious trips.



Telecommunication Infrastructure Technical Requirements



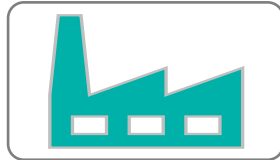
- Backbone telecommunication infrastructure is required to enable extension of control system
- Selection of backbone type are as follows, in order of preference:
 - 1) Fiber Optic
 - 2) DMR with VLAN provision specific for OT system
 - 3) Digital Troposcatter System (DTR) or LEO Satellite, with VLAN provision specific for OT system
- Backup to the main backbone link shall also be provided, e.g.,:
 - ✓ diversified fiber optic route from the main
 - ✓ different FO Metro-E Service Provider
 - ✓ different technology from the main backbone infrastructure (e.g., DMR, DTR, or at minimum VSAT with minimum 2 Mbps bandwidth)
- Remote operation shall not utilize VSAT system as the main backbone communication due to its high latency of 500ms

Telecommunication Infrastructure Technical Requirements

Minimum specification for the selection of backbone telecommunication infrastructure:

**Path Length**

distance between local and remote location

**Availability**

>99.9% (8hr 45min downtime/yr) OR
>99.99% (53min downtime/yr)

**Bandwidth**

minimum 2mbps

**Latency**

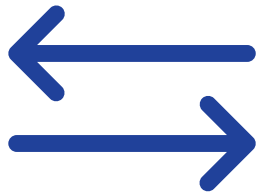
maximum 250ms



14.4 Control System Design Extension Options

How do we extend the local site to remote site ?

There are multiple design options for extension of control system.



Direct Horizontal Extension

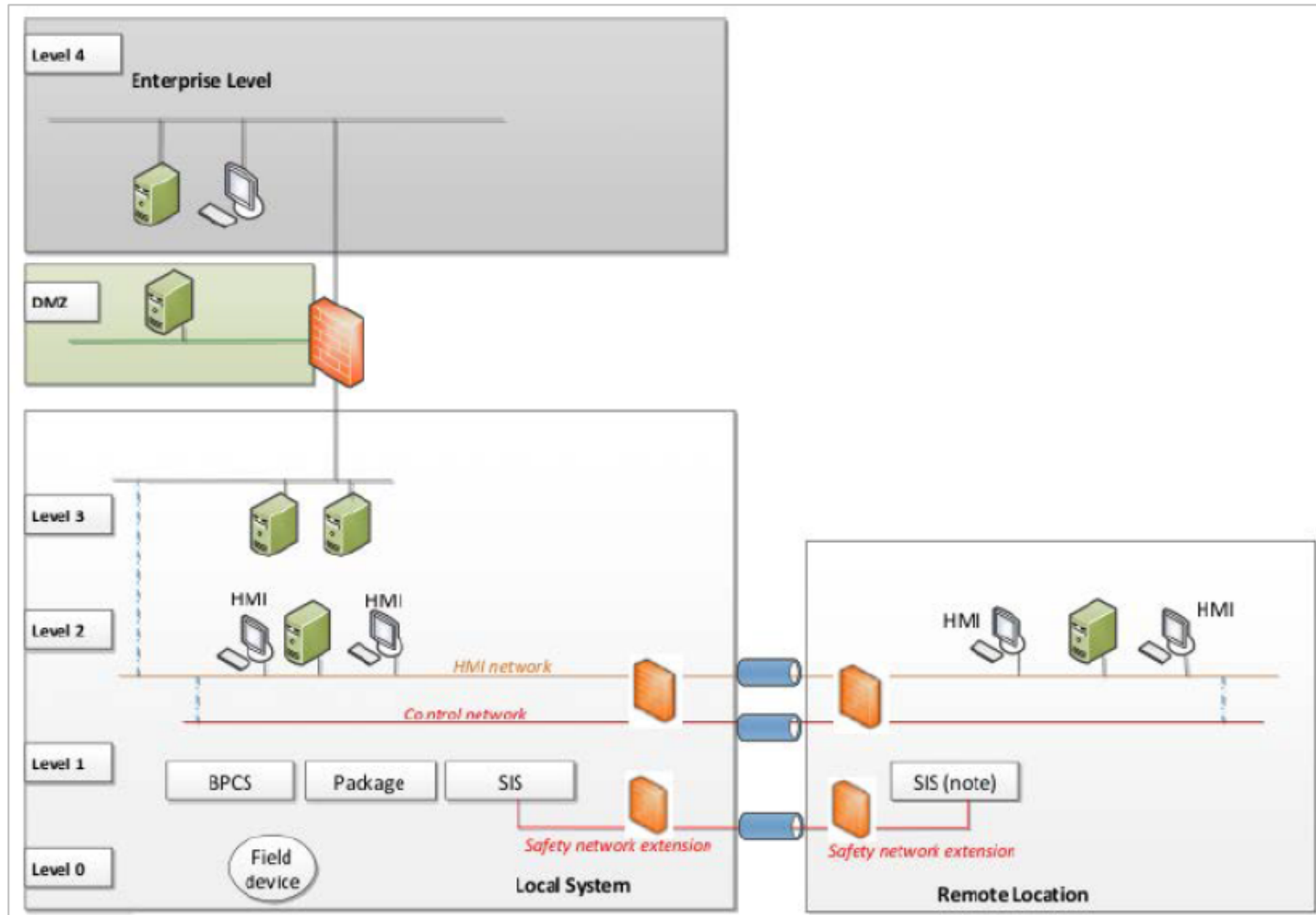


Secure Thin Client Extension



Physically Vertical Extension,
Logically Horizontal Extension

Direct Horizontal Extension via Level 2

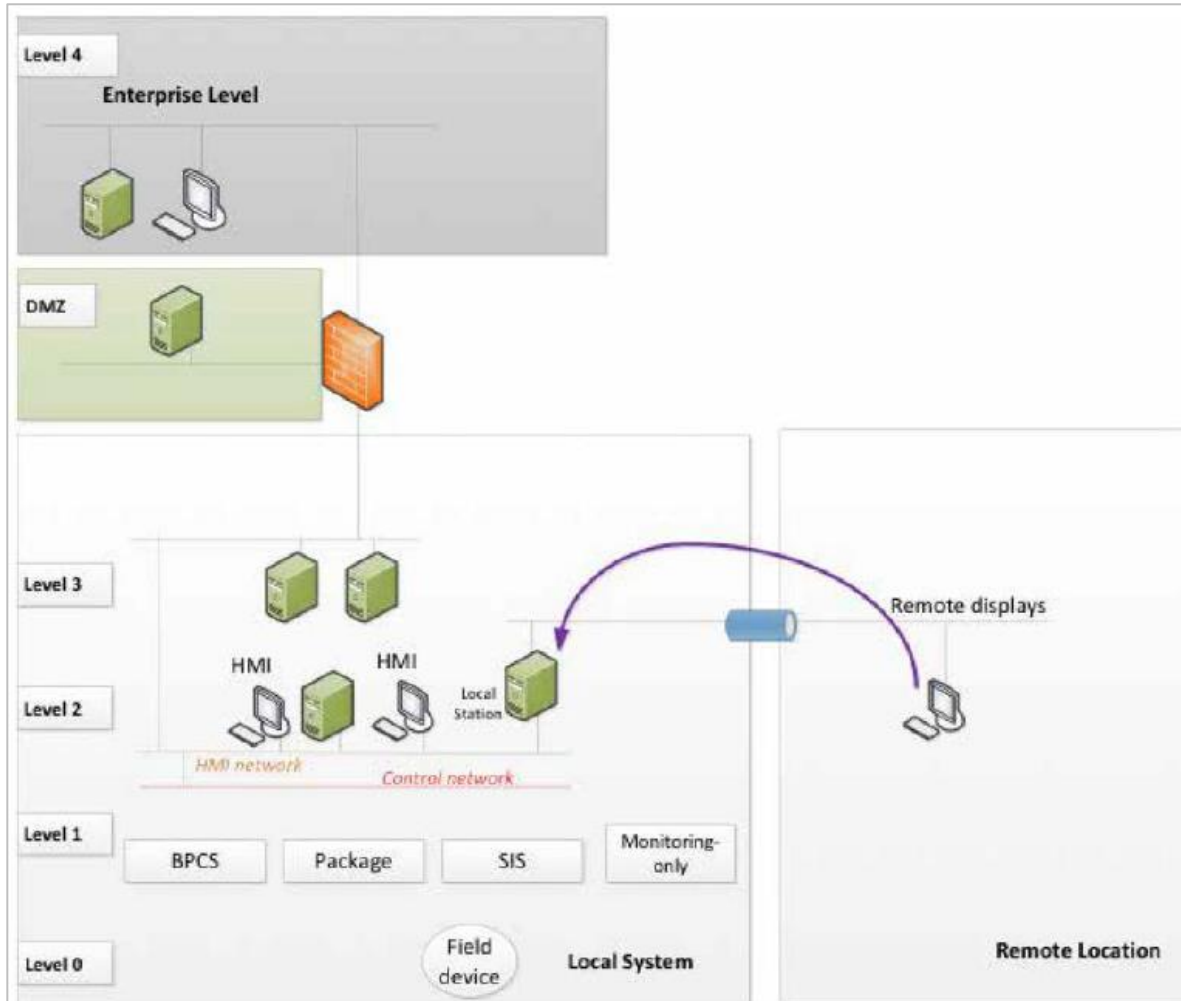


- Level 1 and 2 functions are extended to the remote location using a direct extension of the local system.
- The same communication protocols as the local system network are used for this extension.
- **Low network latency and sufficient bandwidth** are required to support remote users.
- Conduit* for horizontal extension:
 - Physical Network Segregation: dedicated network infrastructure for OT
 - Virtual Network Segregation: shared network infrastructure to virtually segregate between OT network with IT traffic (e.g., VLAN, IP Tunnelling, Encryption)

* Communication channel which is used for extension of the network from local to remote location

Ref: IOGP Report 627, Figure 7: Architecture 1

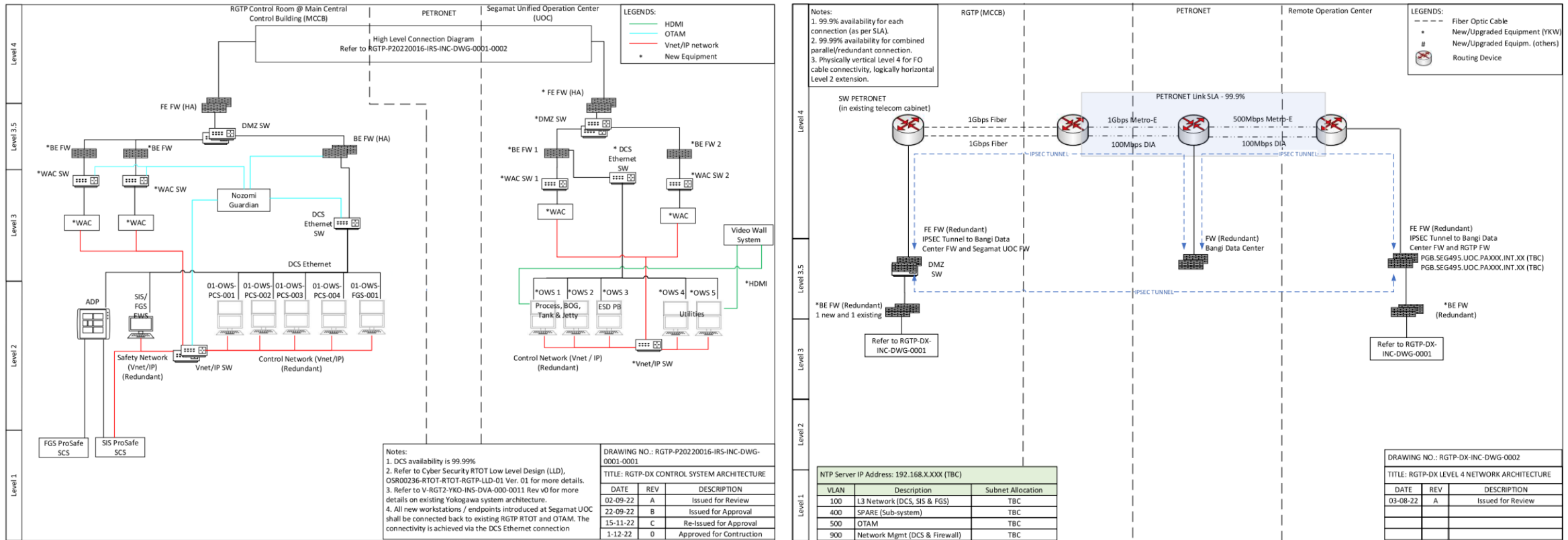
Secure Thin Client Horizontal Extension via Level 2



- Secure Thin Client used to remotely connect to local Operator Workstations using Remote Desktop technologies
- Primary control and Operator Workstations are within the local site
- Remote display access is performed on a dedicated ethernet network
- Recommended to be selected when **existing infrastructure suffers from low bandwidth and/or high network latency.**
- Possible delays in the overall response time

Ref: IOGP Report 627, Figure 8: Architecture 2

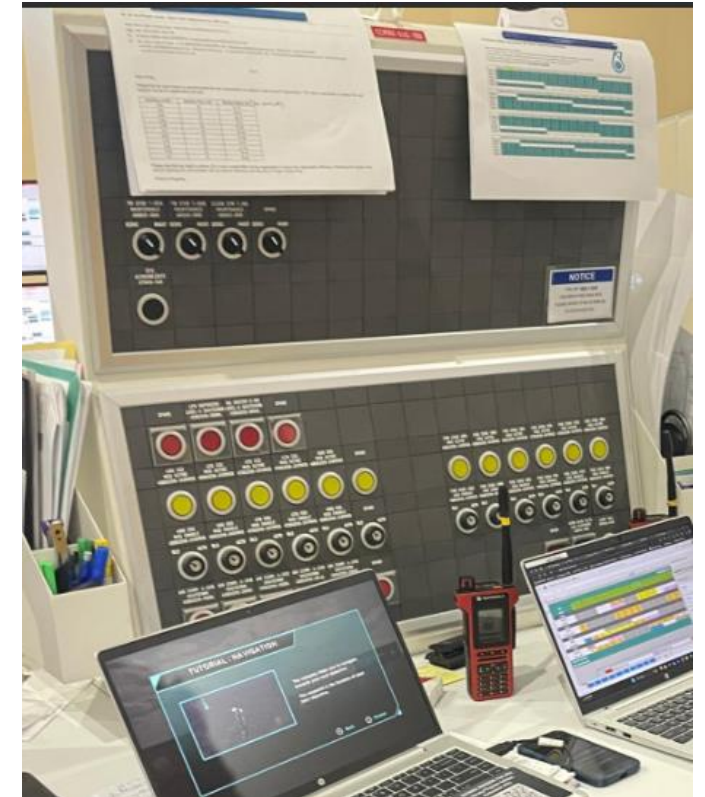
Physically vertical, logically horizontal extension



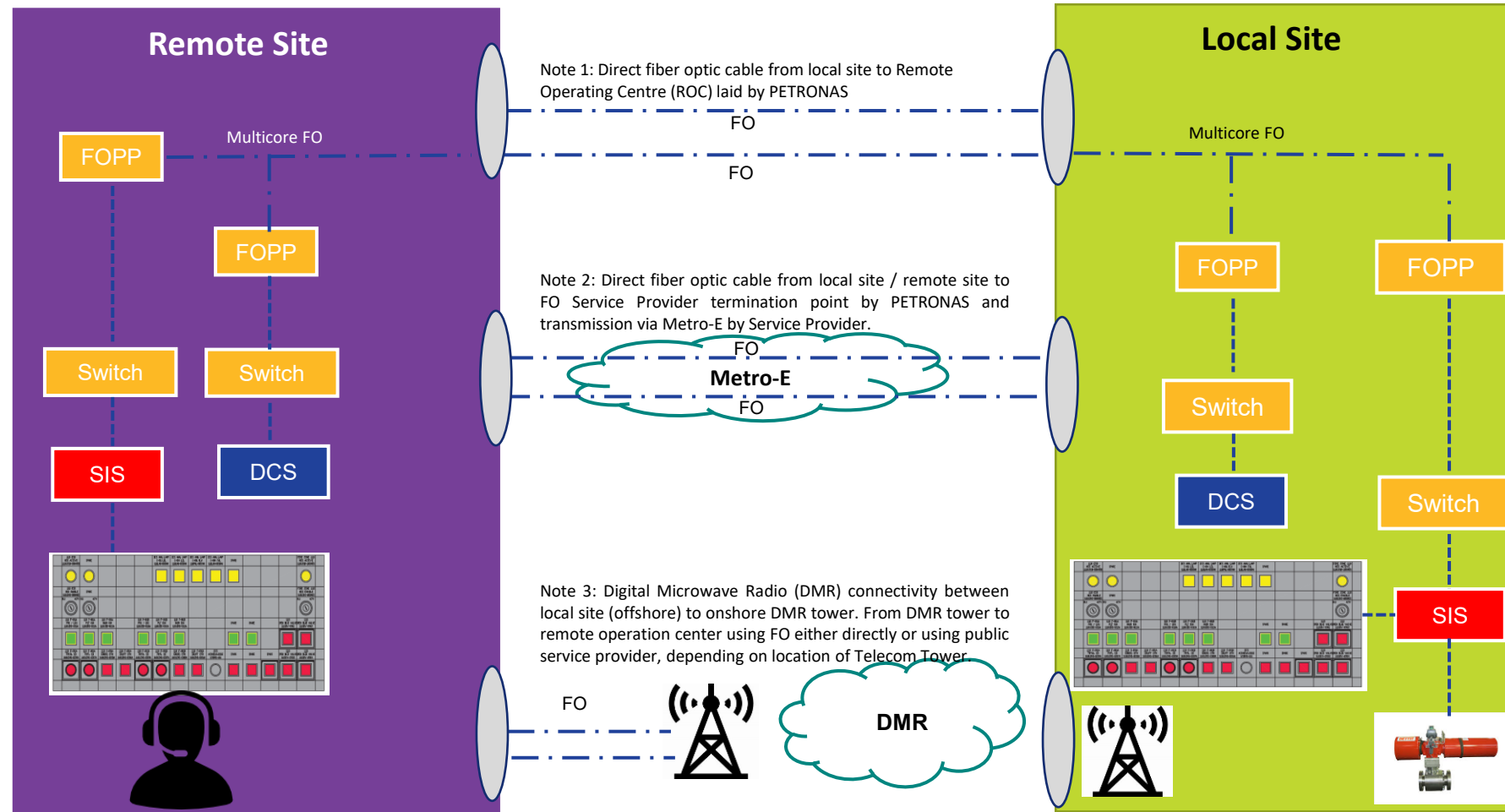
- This extension leverages on existing PETRONET telecommunication infrastructure for vertical connectivity. This is normally done in brownfield facilities to optimize investment in telecom infrastructure. As much as possible, for greenfield facilities, to have dedicated infrastructure for remote operations.
- IPsec Tunnel is required to secure the OT network with addition of back-end and front-end firewall to separate the OT and PETRONET network
- Achieves physical vertical Layer 4 while maintaining logical horizontal Layers 3 and 2.
- Collaboration with PETRONAS Digital Sdn. Bhd. (PDSB) on firewall opening and integration to PETRONAS network is required

Shutdown pushbutton extension

- Local shutdown pushbutton shall be provided (for brownfield facilities, to be maintained) to provide immediate initiation by field personnel (if present) in case of emergency
- Options for remote shutdown pushbutton in ROC:
 - ✓ Hardwired pushbutton on an Auxiliary Panel
 - ✓ Soft pushbutton on HMI
- To ensure continuous production, local safety system shall not be impacted by loss of remote SIS nodes in ROC or soft shutdown pushbutton HMI in ROC, i.e., ROC safety system side shall be non-fail safe.
- This will ensure overall local site availability in the event of any communication issues, especially for offshore links using wireless backbone communication methods such as DMR, DTR, LEO Satellite, or VSAT

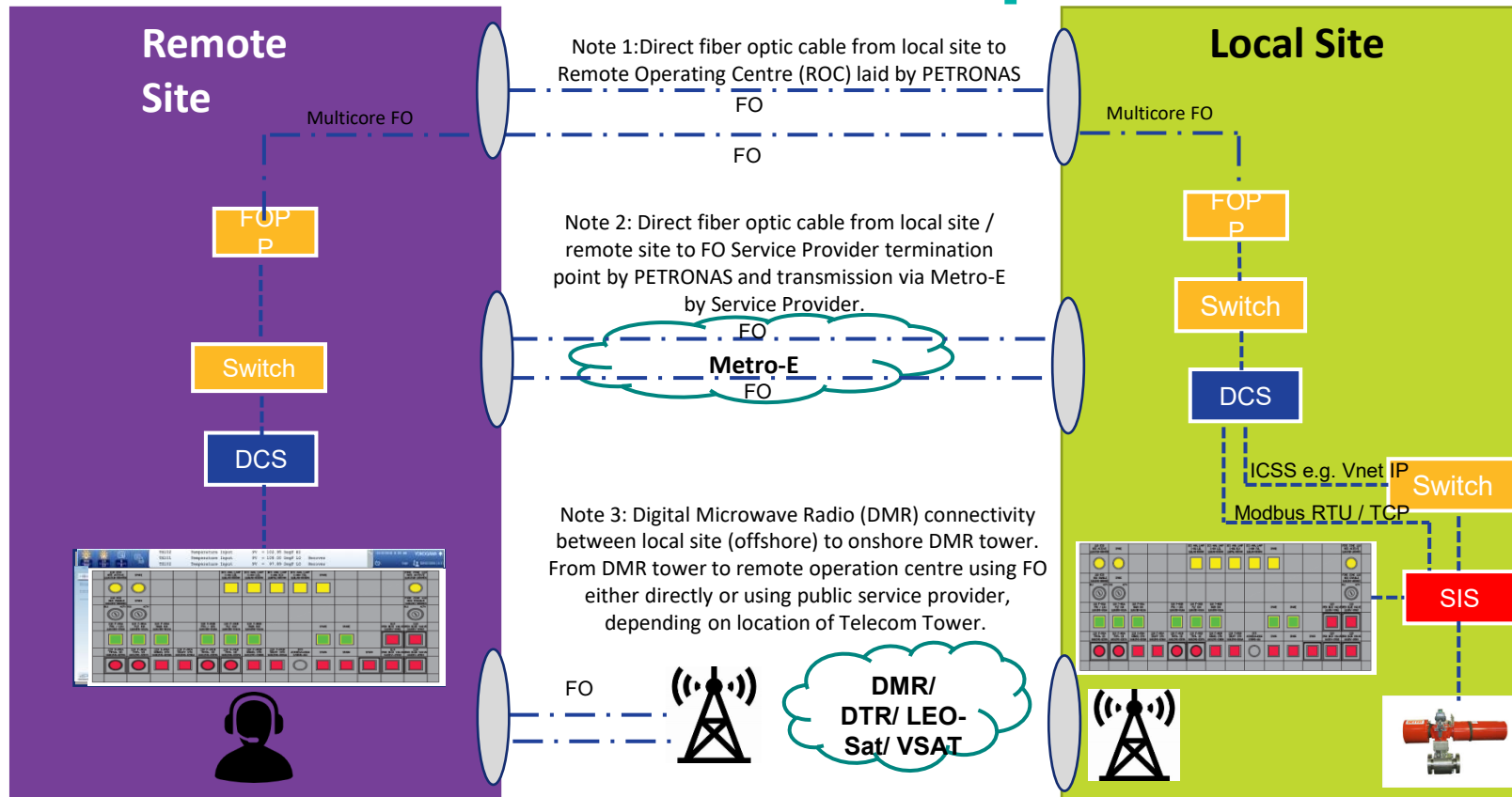


Hardwired Remote Shutdown Pushbutton on an Auxiliary Panel



- The ROC shutdown pushbutton consoles shall be directly connected to the SIS I/O and SIS controller within the ROC cabinet room, which is then connected to the local site SIS via the SIS CSN extension.
- The connection between the shutdown pushbutton Auxiliary Panel and the safety PLC shall have line monitoring.

Soft Remote Shutdown Pushbutton Implemented on an HMI



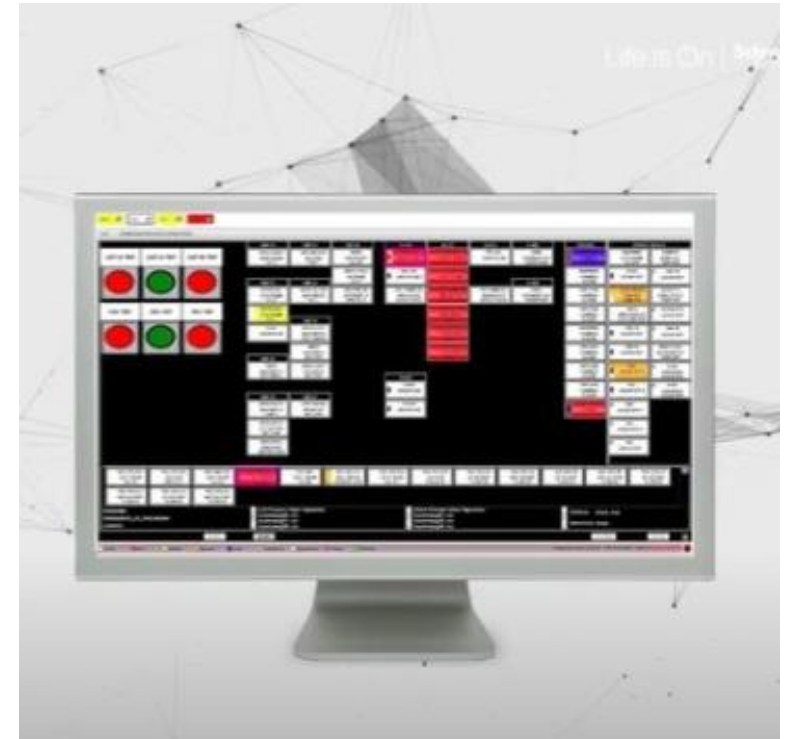
It is possible to implement the shutdown pushbuttons through HMI:

- Recommended to have a dedicated Operator Workstation (OWS) that permanently displays the shutdown pushbutton and alarms HMI, i.e., Panel Operator do not have to toggle/flip any of the pages to access the soft pushbuttons in case of an emergency
- The shutdown pushbutton DCS HMI in ROC will then be connected back to local site DCS network based on the Level 2 DCS horizontal extension.
- At local site, DCS network integration with SIS either via ICSS network (if both using same DCS and SIS e.g. Yokogawa Vnet/IP) or via Modbus RTU/TCP communication to each communication card. Then, the SIS controller will act based on the input from DCS HMI to initiate the action.

Soft Remote Shutdown Pushbutton Implemented on an HMI

If local SIS is not part of ICSS where DCS HMI is not directly integrated with SIS:

- ✓ A non-native HMI or native HMI (with TA approval) can be used at the ROC for the soft pushbutton application.
- ✓ Example: Triconex using Safety View with its proprietary TSAA (Triconex System Access Application) protocol
- ✓ The SIS CSN must be separately extended to the ROC, complying with black channel communication as per IEC61508-2 clause 7.4.11.2.



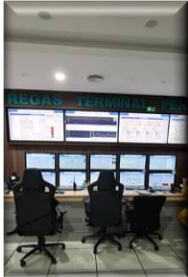
<https://strictlyautomation.wordpress.com>

14.5 Remote Operation Centre (ROC) Design Considerations

Technical requirements for ROC

There are facilities that need to be provisioned for the panel operator in the ROC

The ROC shall be equipped with the following equipment as minimum:



DCS Operator Workstations



Remote shutdown pushbuttons



Radio telecommunications, e.g.,
Radio over IP (RoIP), radio extension



CCTV camera & console



IP telephone



OT network printer

Key considerations in designing a ROC

ROC shall be designed based on the following:

- ✓ ROC control room design shall refer to PTS 14.00.01 Human Factor Engineering – Human / Machine Interface and Control Room Design.
- ✓ Appropriate physical access control shall be implemented at ROC to prevent unauthorized access
- ✓ Proper ingress and egress within the room shall be taken care in managing emergency within the ROC facilities.
- ✓ ROC workstations @ endpoints shall be connected back to the PETRONAS Enterprise Cyber Security solution at local to enable Operating System and Anti-Virus patches update and intrusion monitoring.



14.6 Remote Access, Remote Maintenance and Analytics

New way of working in maintenance

How do we approach maintenance since we have already implemented RAO?



Sources: <https://www.citypng.com/photo/28518/pop-art-man-thinking-blank-bubble-hd-png>

- Do we still need maintenance technician and engineers at site?
- Do we centralize the maintenance team?
- Do we continue Planned Preventive Maintenance (PPM)?

1. We will still need maintenance technicians and engineers, but they will not be at site as generally the maintenance team will be centralized at a location, e.g., ROC or at alternate location that is easily accessible to the local site

2. PPM is time-based and does not achieve cost efficiency requirement of RAO. Hence, with RAO, data-driven maintenance through condition-based, predictive maintenance or prescriptive maintenance is the new way of working. However, the maintenance procedure needs to be updated based on the maintenance strategy selected.

What systems are required to be accessed from ROC to achieve new maintenance approach?



Example of systems:

- DCS/SIS/FGS Engineering Workstation (EWS)
- System1 for machinery performance
- AMADAS EWS
- Electrical Network Monitoring Control System
- Corrosion Monitoring System
- Instrument Asset Maintenance System (iAMS)

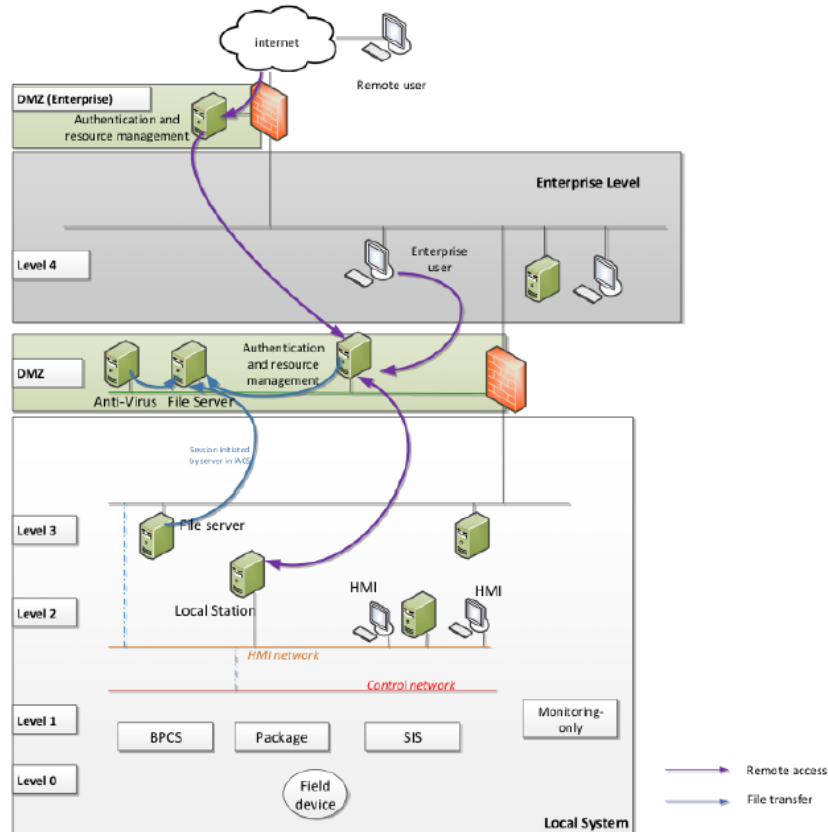
Access can be provided by using Xage remote access

How do we access the system?



IOGP Architecture #6 allows control system endpoints to be remotely accessed from a remote location

5.6. Architecture #6: Remote access from level 4 and above for read and write access

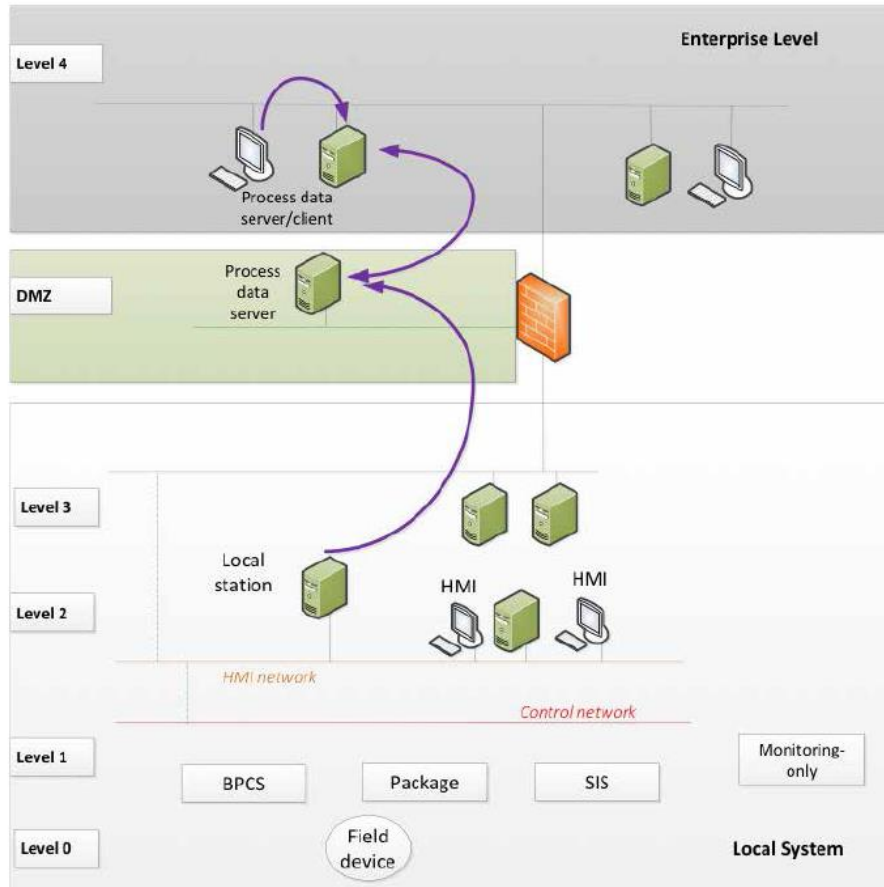


Source: IOGP Report 627 Remote Operation

1. The remote access is enabled via a vertical extension using the enterprise business network, passing through the DMZ level.
2. Specifically designed for remote access of control system Engineering Workstations (EWS) where access is not continuous, seldom and can be configured to be time-bound in accordance with cyber security requirements. It also extends the existing RTOT functionality for OS & AV patch updates using the centralized server.
3. This architecture provides remote access (read & write) and file transfer capabilities utilizing existing network connectivity via Level 4 Enterprise with PETRONET authentications and authorizations.
4. It is suitable for remote maintenance and engineering modifications, such as accessing control system engineering workstations.
5. It also can be used for instrument asset management system (IAMS) remote access to validate field instrument health.

IOGP Architecture #3 is applied for data transfer from L3 Operational Technology (OT) network to L4 Enterprise network

5.3. Architecture #3: Transfer of data to Enterprise level



Source: IOGP Report 627 Remote Operation

1. Data transfer for data analytics, data visualization, data monitoring, production planning, condition monitoring etc. For example, connectivity to Plant Information Management System (PIMS) and Predictive Revitalization of Instrumentation to Maximize Efficiency (PRIME).
2. This architecture only provides view on the data and prevents interactions with the local system.
3. The unidirectional flow of data can be ensured by a network security device such as a data diode or a firewall.

14.7 Advantages and Challenges of Remote Autonomous Operation

The trials and triumphs!

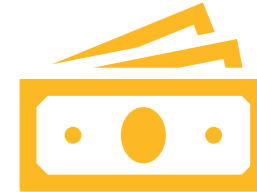
Remote autonomous operation implementation provide substantial advantages that create value



Improve personnel
safety



Increase production
efficiency



Lower operating
costs

However, there are challenges that need to be addressed as part of management of change



Increased cyber
security risks



Workforce
reorganization



Operational
workflow changes

14.8 RAO in Supporting Sustainability

How do we support MFT 0 ?

Remote Operation supports sustainability efforts



**Less carbon
emission** resulting from
reduced traveling

Less energy usage due to
less people at site. E.g.,
HVAC, water



Better work environment
by minimizing exposure
to hazardous areas

Reduced waste with
fewer people present on-
site



Recap – Learning Outcome

By now, you should be able to:

- Explain the different levels of autonomy
- Explain the definition and terminologies used in remote operation
- Describe the remote operation design and remote operation center (ROC) technical requirements
- Describe the remote operation control system extension options
- Describe use cases & application of remote operation for Operation & Maintenance

QUIZ TIME!

Testing your competencies

